VIP | *GAINS*

Simulating Senses: Modelling Personality-Based Perception and Decision Making in NPCs

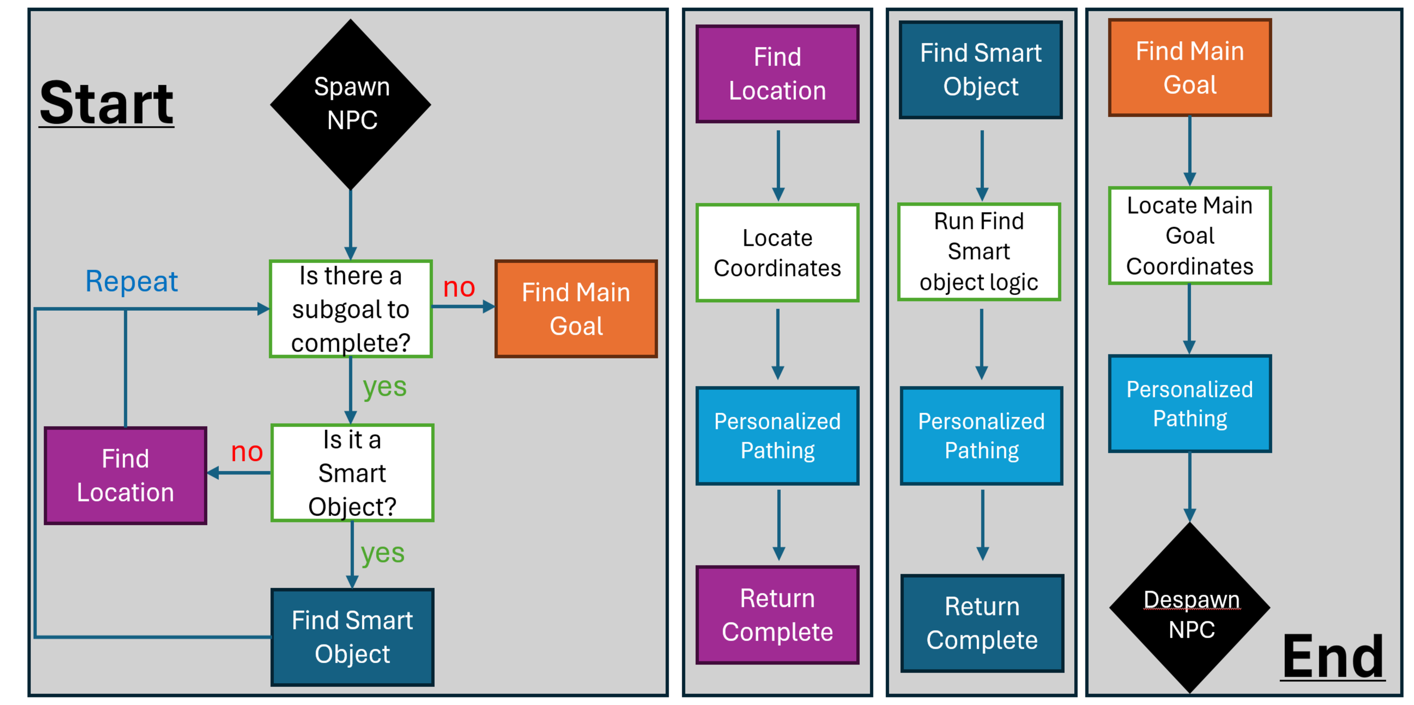
OVERVIEW

This project builds on previous work to utilize the Unreal Engine's AI-driven Non-Player Characters (NPCs) to simulate realistic human behavior and decision-making. This semester, our team focused on equipping NPCs with senses such as vision and hearing, as well as internal traits. Additionally, we focused on creating an environment with interactable elements and subgoals so that NPCs may use these senses and traits, creating emergent behaviors that mimic real-world decision processes without relying on rigid scripting.

This work also has implications for behavioral realism in interior design, urban planning, and crowd modeling, allowing unique insight into human-environment interactions. Ultimately, our work is aiming to bridge the gap between current NPCs who have solely pre-scripted behavior to artificial intelligence-managed NPCs that behave as perceptive, personality-driven individuals in a virtual world.

HIGH-LEVEL OVERVIEW

Our system uses Unreal Engine's Behavior Tree framework to drive NPC actions. Each NPC is controlled by an AI Controller that updates a shared Blackboard, allowing behaviors to react dynamically to changing inputs. The scene is set up with both static and dynamic elements (such as obstacles, sound emitters, and goal locations). Each element provides sensory information for the NPCs to interpret. NPCs rely on two modes of perception–sight and hearing–along with internal preferences and sub-goals. Together, these components allow NPCs to make autonomous decisions that reflect environmental cues and personality-based traits.



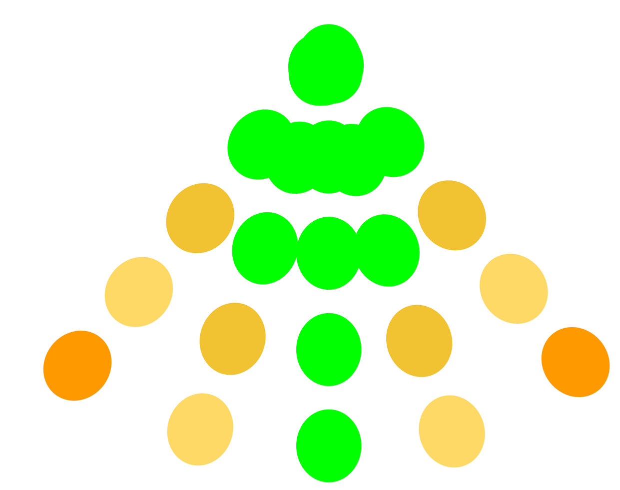
A high-level flow of the NPC system, showing how goals, sub-goals, and personalized pathing guide decision-making throughout the behavior tree.

SUBGOALS

Our goal with being able to process sub-goals in our NPCs is to simulate more realistic behavior by allowing our NPCs to take individualized decisions on their pathing toward their final destination. The addition of this feature allows our NPCs to process other interactive elements like our interactable objects mentioned previously, before completing their main goal. How it functions is our NPCs will process sub goals (if any) they must complete in their behavior tree before attempting to complete their main task. This incorporates the aforementioned hearing and sight perception for our NPCs as the pathing toward mini-goals is dependent on each individual NPC's sense and preferences related to environmental factors.

PERCEPTION: SIGHT

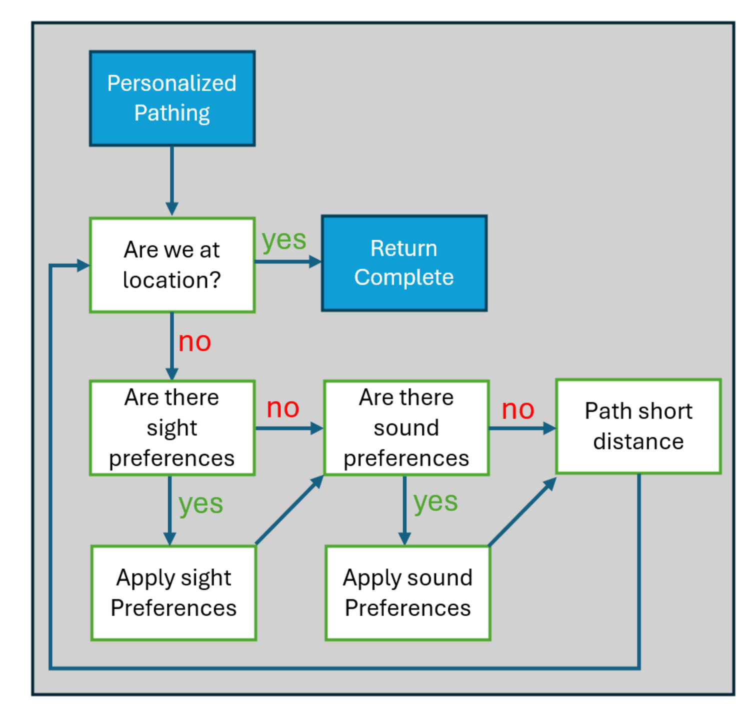
NPC sight cone, based on distance and angle



Sight lets NPCs perceive favorable versus unfavorable situations and navigate accordingly. The core logic implements a decision step that checks whether there is a line of sight to the target. We express vision entirely through Unreal's Environmental Query System (EQS). This EQS query evaluates candidate points within a range based on distance and angle (peripheral vs. direct). The best result updates a blackboard key read by the Behavior Tree to steer movement toward what the NPC can see. This system allows our NPCs to simulate human sight in virtual environments, allowing them to navigate realistically.

PERCEPTION: HEARING

Our goal was to simulate human-like variability in how NPCs respond to noise—some are drawn towards it while others avoid it. Using Unreal Engine's AI Perception system, NPCs detect sound events in their environment. When a sound is perceived, the AI Controller updates relevant blackboard variables. The Behavior Tree then combines these with each NPC's noise preference trait to decide whether to move toward or away from the sound while pursuing its current goal.



NPC pathing guided by sight and hearing perception

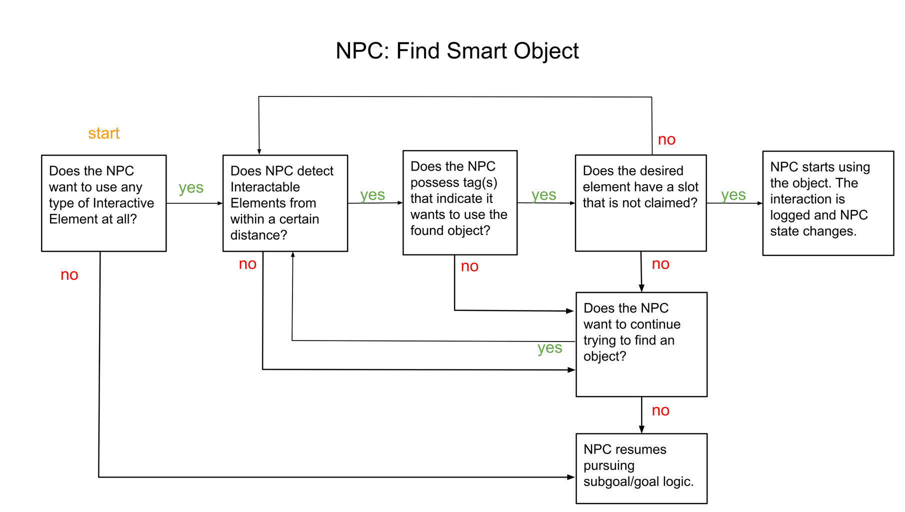
REFLECTION

Overall, the integration of sight and hearing perception in NPCs, as well as the creation of an interactive environment with subgoals and objects, have made this project one step closer to achieving realistic simulation of human behavior which is not driven by rigid scripting.

Future extensions to this framework include developing other senses like touch, more complex personality-driven interactions beyond simple preference, and interaction between NPCs. Additionally, testing methods must be developed to evaluate the simulated NPC-environment interactions and their usefulness in enhancing understandings in crowd modeling, architecture, and other fields.

INTERACTIVE ELEMENTS

Including NPC interaction with everyday objects such as furniture is crucial in working towards simulated human behavior. The Interactive Elements system uses an experimental feature in Unreal known as Smart Objects to represent seats, landmarks, and water fountains, and more. The flow chart to the right summarizes the logic for how the NPC chooses to find, use, or give up on finding objects depending on object availability and NPC tags. Integrating interactable elements to the project makes it one step closer to simulating realistic human behavior and moment-to-moment decision-making in a virtual environment.



If the NPC wants to use any object, it first looks for objects within a certain range. If the object has an available slot (it is not already being used) and the NPC has a tag indicating that it wants to use it, the NPC moves to the slot, and the desired behavior is executed and recorded. Otherwise, the NPC chooses to either continue looking for interactable elements or pursue another goal.

PROJECT NOTES

This project is a continuation of past semesters' work by the GAINS (Game-Dev AI NPC Simulation) team, led by Professor David Barbarash, as part of the VIP (Vertically Integrated Projects) course.

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